STAT 526 HW 9

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Problem 0

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Problem 1

To assess the gender risk of death from coronary heart disease (CHD), a study was performed controlling for genetic and environmental factors by using fraternal twins, one male and one female. The age at which a male/female twin died of CHD was recorded. The dataset twin contains the following samples from this study.

a.)

Calculate the Kaplan-Meier estimates of the survival functions for the male group.

Our estimate function is:

$$\hat{S}(t) = \prod_{i:t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

We can make a table for that:

Age (t_i)	At Risk (n_i)	Deaths (d_i)	Censored	$1 - d_i/n_i$	$\hat{S}(t)$
0	12	0	0	-	1.000
49	12	1	0	0.917	0.917
50	11	0	1	1.000	0.917
56	10	1	0	0.900	0.825
61	9	0	1	1.000	0.825
67	8	0	1	1.000	0.825
68	7	0	1	1.000	0.825
69	6	1	0	0.833	0.688
70	5	1	0	0.800	0.550
74	4	2	0	0.500	0.275
75	2	1	0	0.500	0.138
81	1	1	0	0.000	0.000

b.)

> Using the output below, state the hypothesis that gender is a risk factor of death from CHD, and test the hypothesis while controlling for genetic and environmental factors.

This is Log-Rank Test of homogeneity.

The null hypothesis is that there is no differences between the survival probability between the male group and female group.

The alternative hypothesis there is a significant difference between the survival probability between the male group and female group.

We can do the test. From the output, we observe 7 female deaths and 8 male deaths with expected deaths to be 7.91 for female and 7.09 for male.

We need to calculate the test statistic. We first solve for variance given our output:

$$\frac{(7-7.91)^2}{V} = 0.299 \implies v = 2.76957$$

We can calculate the test statistic $((7-7.91)^2 + (8-7.09)^2)/\sqrt(2.76957)$ or simply use the R output of 0.299 on 1 degree of freedom.

```r
pchisq(0.299, 1, lower.tail = FALSE)
```

```
## [1] 0.58451
```

For $\chi^2 = 0.299$ with 1 df, the p-value is approximately 0.5845.

Since p-value is greater than confidence level 0.05, we fail to reject the null hypothesis. We believe that there is no significant evidence that the two groups have different survival probabilities.

c.)

> An accelerated failure time model is fitted to the data as shown below. Using the output below, predict the probability for a female with CHD to survive past 75.

Survival function is

$$S(t) = \exp\left(-t \cdot \lambda\right) = \exp\left(-t \cdot e^{-\mu}\right) = \exp(-t \cdot e^{-(\beta_0 + \beta_1 x)})$$

Female is baseline so $\mu = \text{Intercept} = 4.772$.

The scale parameter is $\lambda = e^{-\mu} = e^{-4.772} \approx 0.008463$. Then:

$$S(75) = \exp(-75 \cdot 0.008463) = \exp(-0.634725) \approx 0.530$$

The predicted survival probability to survive past age 75 is around 53%.

d.)

> A proportional hazards model is fitted to the data as shown below. For a twin consisting of a male and female, what is the effect of gender on the survival?

We see outputs coef (β) = 0.26 and exp(coef) (HR) = 1.30. The exp(coef) is the hazard ratio meaning we have 30% higher risk of death from CHD as a male compared to female.

However, the p-value is 0.62 and our confidence interval includes 1, meaning we do not have significant evidence of the effect of gender on survival.

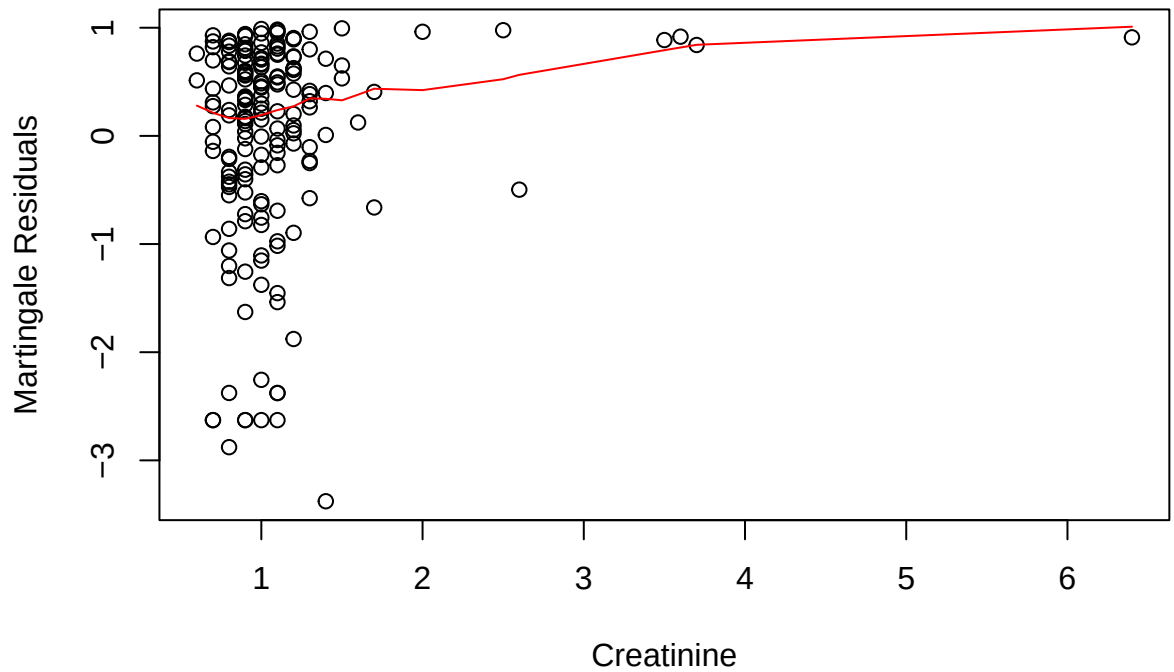
Problem 2

Consider the dataset mgus in the library survival (use ?mgus for information on the dataset). Your goal is to model the survival time of these patients using Cox proportional hazard model (ignore variables pcdx and pctime, which have a large proportion of missing data). Verify the plausibility of a proportional hazard model for this dataset. Select important predictors in the model, and their functional form. Evaluate the quality of fit (optional). Interpret model parameters and discuss the results. Hint: i) logarithm transformation of creat is suggested due to its high skewness ii) please use resid() function to obtain martingale residuals. Use visual diagnosis of martingale residuals (under null model) vs predictor. iii) To diagnose proportional hazard assumption with respect to continuous predictor, you can convert this continuous predictor to categorical variable (say partition based on empirical quantity), then use the plot of log(-logS(t)) vs log(t).

```r
library(survival)
library(survminer)
```

```
## Warning: package 'survminer' was built under R version 4.5.3

## Loading required package: ggplot2

## Loading required package: ggpubr

## Warning: package 'ggpubr' was built under R version 4.5.3

##
## Attaching package: 'survminer'

## The following object is masked from 'package:survival':
##
##     myeloma
```

```r
# Ignore pcdx and pctime as per instructions
df <- mgus[, !(names(mgus) %in% c("pcdx", "pctime"))]
df <- na.omit(df) # get rid of NA

# Null model to obtain martingale residuals
null_model <- coxph(Surv(futime, death) ~ 1, data = df)
df$mart_resid <- resid(null_model, type = "martingale")

# Visual diagnosis of martingale residuals (under null model) vs predictor by hint (ii)
plot(df$creat, df$mart_resid, xlab = "Creatinine", ylab = "Martingale Residuals")
lines(lowess(df$creat, df$mart_resid), col = "red")
```

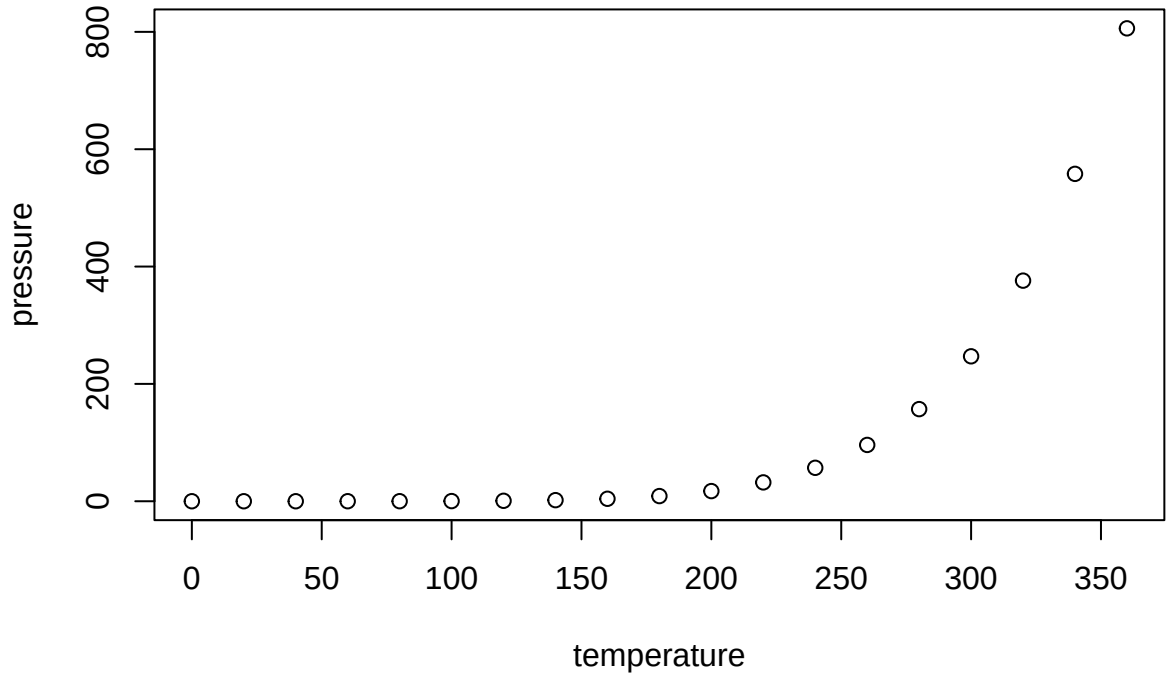



```
# Transform creat as suggested by hint (i)
df$log_creat <- log(df$creat)
```

```
library(survival)
library(survminer)
```

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.